



DIABETES MELLITUS

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OBJECTIVES

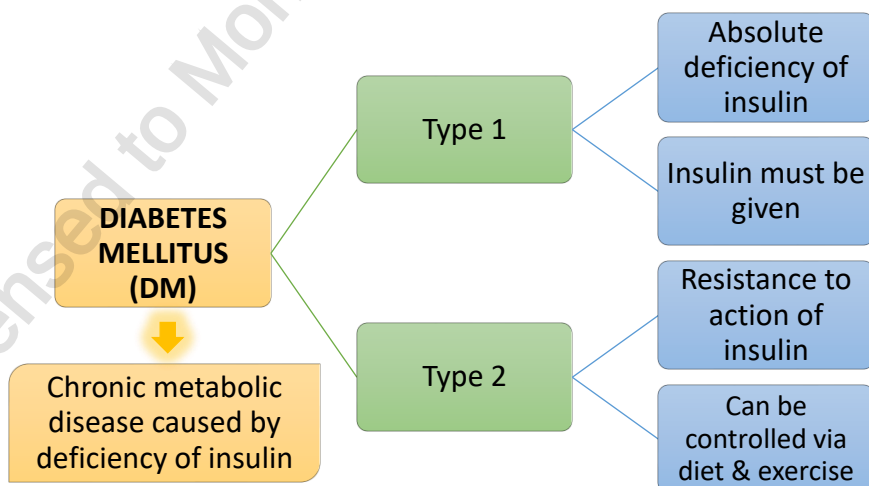
To be able to do the following with respect to diabetes mellitus:

- Compare and contrast type 1 and type 2
- Understand clinical course and diagnostic testing
- Determine causative factors
- Identify clinical manifestations and potential complications
- Outline nursing interventions

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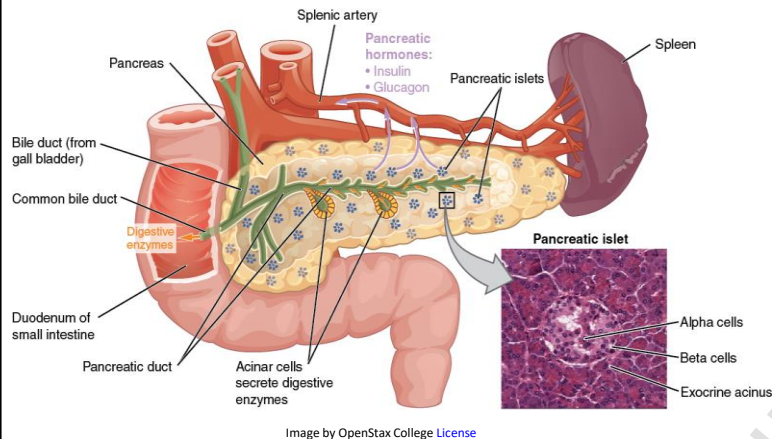
DEFINITIONS



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PATHOPHYSIOLOGY



Pancreas:

- Involves secretion of insulin (produced by beta cells) and glucagon (produced by alpha cells), which regulate the rate of glucose metabolism in the body
- DM occurs when the pancreas is unable to produce insulin or insulin cell receptors are resistant
- Without insulin, glucose accumulates in the body → hyperglycemia (increased blood glucose levels) & classic symptoms of diabetes mellitus

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PATHOPHYSIOLOGY

BODY'S RESPONSE TO INSULIN DEFICIENCY	DEFINITIONS
Decreased glycogenesis	Conversion of glucose to glycogen
Increased glycogenolysis	Conversion of glycogen to glucose
Increased gluconeogenesis	Formation of glucose
Increased lipolysis	Breakdown of triglycerides to glycerol and free fatty acids
Increased ketogenesis	Formation of ketones from free fatty acids
Proteolysis	Breakdown of proteins with amino acid release in muscles

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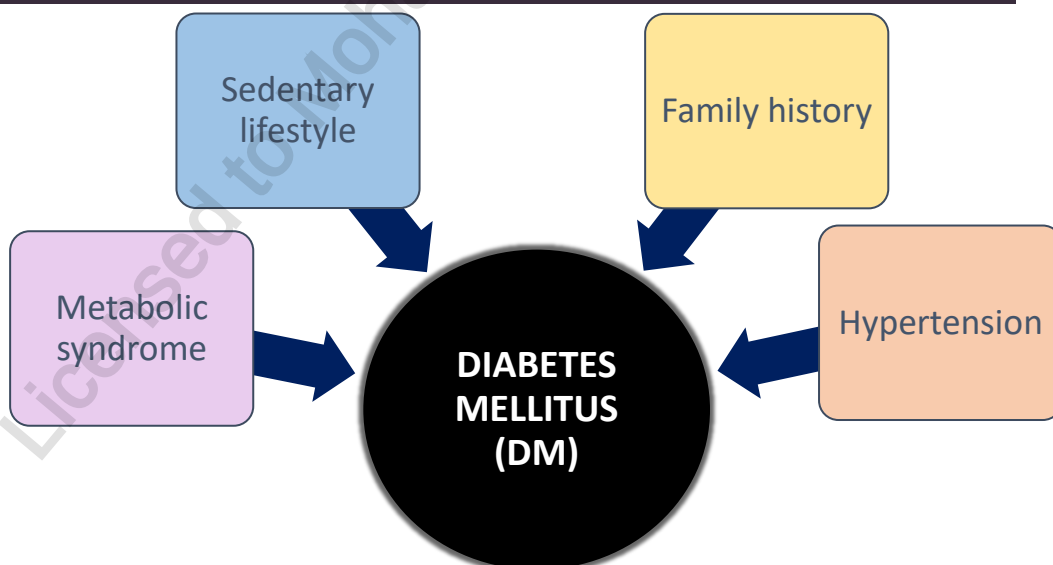
PATHOPHYSIOLOGY – DM TYPE 1 & 2

Characteristics	Type I	Type II
Other names	Insulin-dependent DM	Non-insulin-dependent DM
% of diabetics	5-10%	90%
Age at onset	Usually <30 years Peaks at 12-14 years	Usually >40 years
Pancreatic function	Usually none	Insulin present in normal, low, or high amounts
Pathogenesis	Insulin absent	Insulin present but defect in insulin secretion Tissue resistance to insulin
Obesity	Uncommon	Common (60-90%)
Ketoacidosis	Often present	Rare

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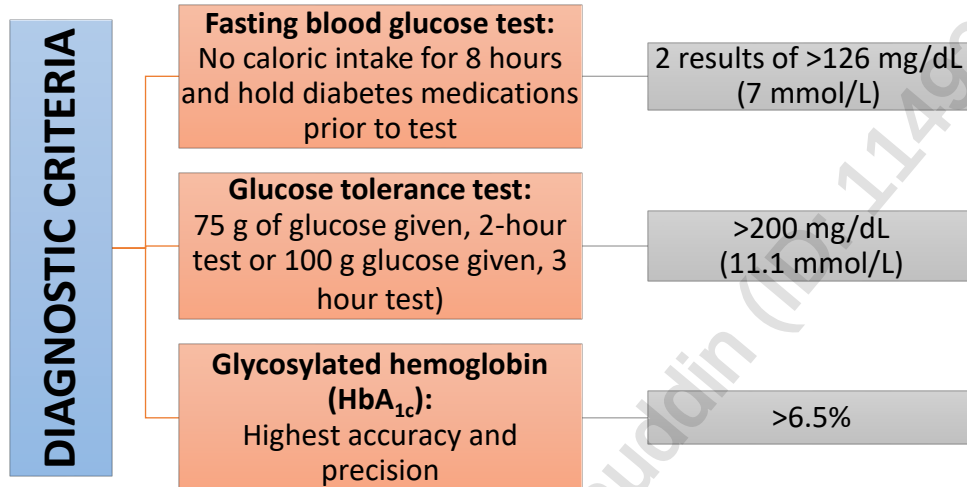
ETIOLOGY – RISK FACTORS



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DIAGNOSIS

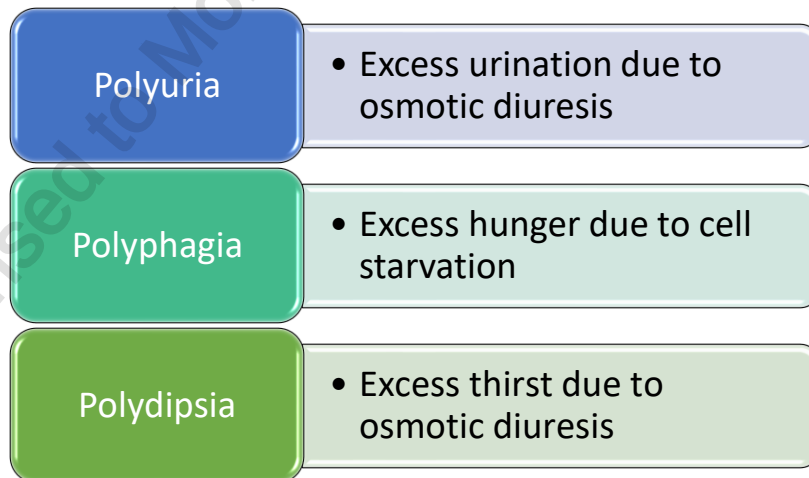


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CLINICAL MANIFESTATIONS

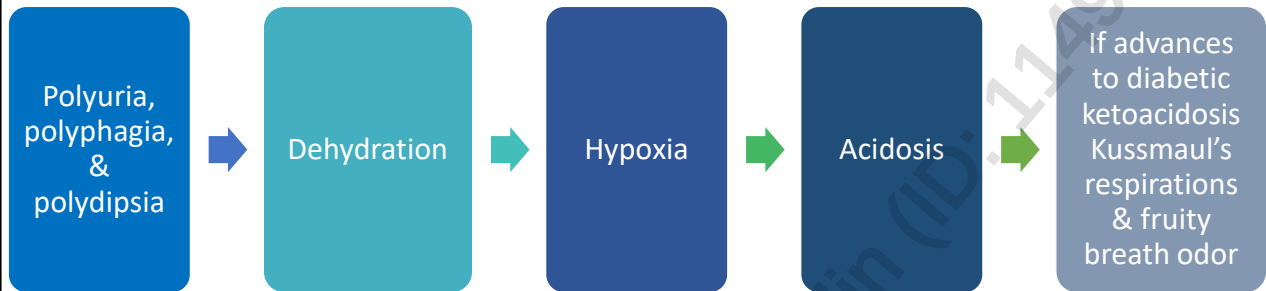
CLASSIC SYMPTOMS OF DM



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CLINICAL MANIFESTATIONS

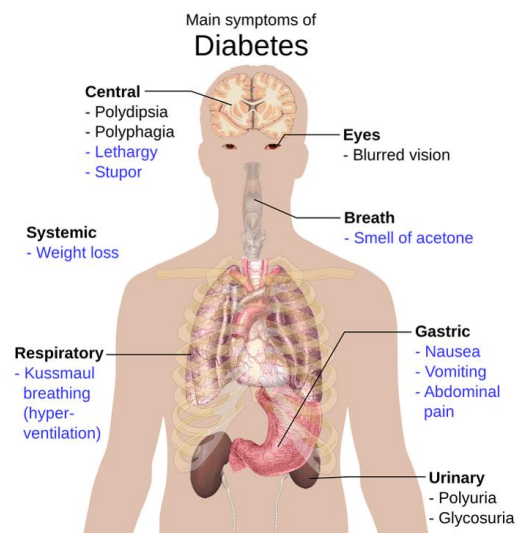


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CLINICAL MANIFESTATIONS

- With insulin deficiency, fats break down, releasing free fatty acids
- Conversion of fatty acids to ketone bodies occurs, leading to metabolic acidosis
- Hyperglycemia: increased thirst, headaches, blurred vision, frequent urination, fatigue, weight loss, and blood sugar more than 200 mg/dL (11.1 mmol/L)
- Hypoglycemia: diaphoresis, chills, and clamminess, and blood sugar below 70 mg/dL (3.9 mmol/L)



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COMPLICATIONS – HYPOGLYCEMIA

- *Occurs when blood glucose is 70 mg/dL (3.9 mmol/L)*
- **Mild hypoglycemia:**
Administer 15 g of fast-acting simple carbohydrate & recheck glucose level in 15 minutes. A snack with protein and carbohydrates should be administered unless client eats regular meal within 60 minutes
- **Moderate hypoglycemia:**
Administer 15-30 g of fast-acting simple carbohydrate as well as more food after 10-15 minutes
- **Severe hypoglycemia:**
If client is unconscious, administer glucagon IM or SC and give second dose in 10 minutes if client still remains unconscious. Small meal given after client is awake. In emergencies, D50W used.



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COMPLICATIONS – DIABETIC KETOACIDOSIS

- Life threatening complication of type 1 DM due to decreased insulin
- Results in hyperglycemia, dehydration, ketosis, Kussmaul's respirations, fruity breath, and acidosis
- Intervene by restoring blood volume, addressing dehydration with 0.9% normal saline, treating hyperglycemia with insulin, monitoring potassium levels, and administering IV bolus of insulin 5-10 units

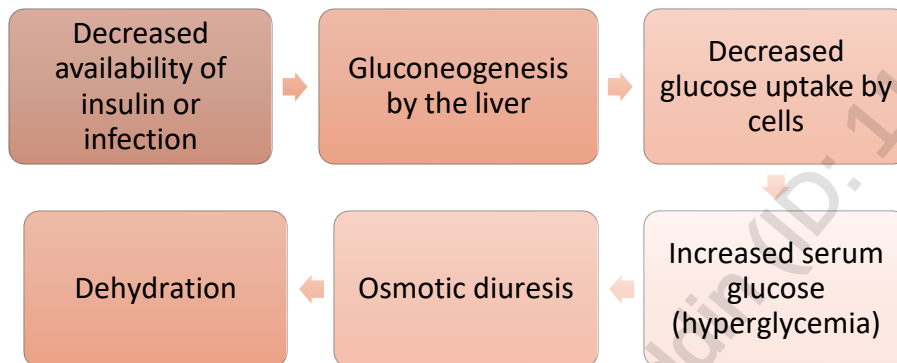


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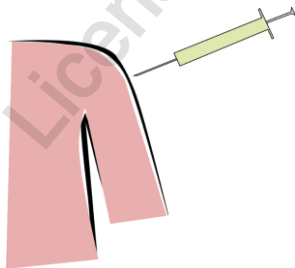
COMPLICATIONS – DIABETIC KETOACIDOSIS



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COMPLICATIONS – HHNS



Hyperglycemic Hyperosmolar Nonketotic Syndrome:

- Hyperglycemia that occurs without ketosis and acidosis as there is enough insulin to prevent breakdown of fats for energy
- Typically seen in clients with DM2
- Treatment includes insulin administration, fluid replacement, and correction of electrolyte imbalances

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COMPLICATIONS – DKA VERSUS HHNS

Findings	Diabetic Ketoacidosis (DKA)	Hyperglycemia Hyperosmolar Nonketotic Syndrome (HHNS)
Onset	Sudden	Gradual
Presentation	Ketosis, Kussmaul breathing, “fruity” breath, dehydration, electrolyte loss	Neurological symptoms, dehydration, electrolyte loss
Serum Glucose	>250 mg/dL (13.9 mmol/L)	>600 mg/dL (33.33 mmol/L)
Serum Ketones	Positive	Negative
Serum pH	<7.35	>7.35
Urine Ketones	Positive	Negative

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COMPLICATIONS – MACROVASCULAR

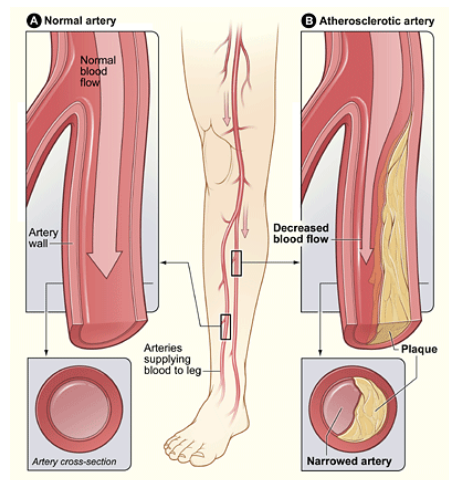
Coronary artery disease

Cardiomyopathy

Hypertension

Cerebrovascular disease

Peripheral vascular disease



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COMPLICATIONS – MICROVASCULAR

Retinopathy

Nephropathy

Neuropathy

Word cloud containing terms related to diabetes and its complications: PANCREAS, CELLS, RESISTANCE, METABOLISM, GLUCOSE, STAGES, ISLETS, MONITOR, WEIGHT, CONGENITAL, NERVE, HEALTHCARE, SYMPTOMS, TYPE, COMPLICATIONS, HYPERGLYCEMIA, SUGAR, RESPOND, INSULIN, MELLITUS, PANCREAS, KETOACIDOSIS, INJECT, ADULTS, ENDOCRINE, ACUTE, WELL, INJECT, SENSITIVITY, TYPE, HEALTHCARE, SYMPTOMS, COMPLICATIONS.

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COMPLICATIONS – RETINOPATHY

Rupture of micro-aneurysms in retina

Sudden blurred vision

Gradual loss of vision due to retinal detachment

Cataracts due to opacity of lens

Manage hypertension & blood sugar

Photocoagulation (laser therapy) to decrease scarring

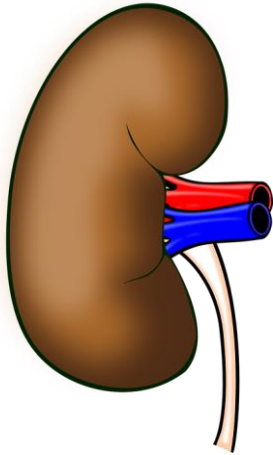
Vitrectomy to decrease tension on retina

Cataract removal to improve vision

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COMPLICATIONS – NEPHROPATHY



Clinical Manifestations:

- Microalbuminuria
- Fatigue
- Thirst
- Anemia
- Weight loss
- Frequent urinary tract infection (UTI)

Interventions:

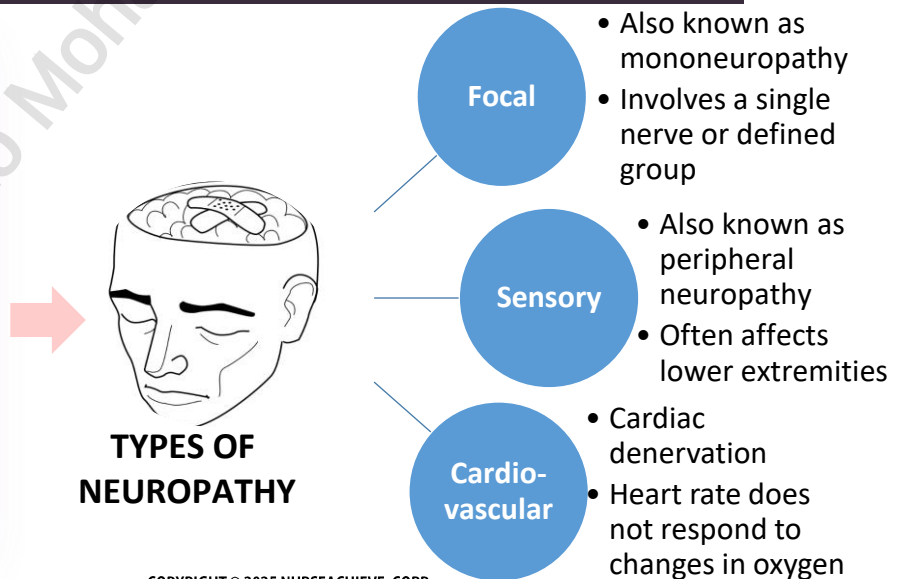
- Manage hypertension and blood glucose levels
- Monitor blood urea nitrogen, creatinine, and urine albumin
- Restrict dietary protein, sodium, and potassium
- Consider dialysis as well as kidney and pancreas transplantation

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COMPLICATIONS – NEUROPATHY

- Deterioration of nervous system
- Complications include foot ulcers, gastric paresis, and erectile dysfunction
- Location of neuropathy dictates manifestations



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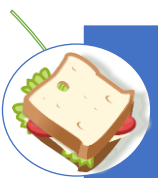
GOALS OF CARE



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INTERVENTIONS – NONPHARMACOLOGICAL



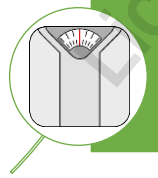
Diet:

- Fats: 20-35%
- Protein and Carbohydrate: Individualized for each client
- Alcohol: ~7 drink/week (F), ~14 drink/week (M)



Exercise:

- At least 150 min/week of moderate intensity exercise



Weight Loss:

- Weight loss of 5-7% of initial body weight improves insulin sensitivity and levels of glucose

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INTERVENTIONS – PHARMACOLOGICAL

Insulin: directly stimulates glucose uptake by the cells

Use: replacement therapy in type 1 diabetes, may be indicated in type 2 diabetes if lifestyle and antidiabetics are not effective

Side effects:

- Hypoglycemia (tachycardia, confusion, sweating, drowsiness)
- Lipodystrophy and injection site pain
- Weight gain

Bolus/Mealtime Insulin: to mimic natural insulin secretion in response to a meal

Basal/Background Insulin: to control insulin levels between meals and overnight



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INTERVENTIONS – TYPES OF INSULIN

Insulin	RAPID	REGULAR	INTERMEDIATE	LONG - ACTING
	Lispro (Humalog) Aspart (Novolog) Glulisine (Apidra)	Humulin/ Novolin R	Humulin NPH Novolin ge NPH	Detemir (Levemir) Glargine (Lantus)
Onset	10 – 30 min	30 – 60 min	1.5 – 4 h	1 – 2 h
Peak	30 min – 3 h	2 – 5 h	4 – 12 h	Less defined or no peak
Duration	3 – 5 h	5 – 8 h	12 – 18 h	16 – 24 h

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INTERVENTIONS – TYPES OF INSULIN

Insulin	ULTRA-RAPID	REGULAR	INTERMEDIATE	LONG - ACTING	
	Lispro Aspart Glulisine	Humulin/ Novolin R	Humulin/ Novolin NPH	Detemir Glargine	
Basal/Bolus	Bolus	Bolus	Basal	Basal	Basal
Appearance	Clear	Clear	Cloudy	Clear	Clear
Comments	• Less Hypoglycemia than regular insulin • Take right before eating	Take 30 min prior to eating	Given at bedtime or BID	• Peakless but Detemir seems to have a faster peak and shorter duration compared to Glargine • Usually daily dosing but Detemir given bid	

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INTERVENTIONS - INSULIN ADMINISTRATION

DRAWING UP AND MIXING INSULIN (regular and intermediate/NPH)

1. Wash hands
2. Rotate NPH insulin bottle to mix
3. Inspect the insulin bottle- proper type, concentration, expiration date, cap integrity, color
4. Sterilize insulin vial caps
5. Draw air into syringe equal to total dosage
6. Inject air into NPH (CLOUDY) vial, equal to NPH dose
7. Inject air into regular (CLEAR) vial, equal to regular dose
8. Withdraw regular insulin dose
9. Withdraw NPH dose

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INTERVENTIONS - INSULIN ADMINISTRATION

Syringe/Needle Selection

- 25-27-gauge, 0.5 inch
- Insulin pens
- U-100 specific syringes (0.3, 0.5, and 1.0 mL sizes)

Insulin injection sites:

- The abdomen is the preferred injection site- this is the fastest absorption site, followed by the arm, thigh, and buttock

Insulin Storage:

- Cartridges for insulin pens can be stored at room temperature for 28 days, or in the refrigerator if not in use
- Mixed insulins can be stored in refrigerator for 4 weeks

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INTERVENTIONS - COMPLICATIONS

Somogyi effect: rebound morning hyperglycemia occurs due to a release of counterregulatory hormones, as a result of decline in blood glucose levels overnight (between 2 and 4 am). This condition can pose a risk to clients, as signs of hyperglycemia may prompt an increase in insulin dose, resulting in rapid rebound hypoglycemia

Treatment and Prevention:

- Bedtime snacks
- **Reduction** in insulin dose

Dawn Phenomenon: hyperglycemia on awakening due to increased secretion of cortisol and growth hormone throughout the night

Treatment and prevention:

- **Increase** insulin dose
- Adjust administration timing

*it is important to differentiate between these two conditions to initiate the correct treatment. High blood glucose between 2 and 4 am suggest dawn phenomenon.

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INTERVENTIONS - ANTIDIABETIC DRUGS

DRUG NAME	MECHANISM	SIDE EFFECTS
Sulfonylureas	stimulates insulin release and increases insulin sensitivity	- Hypoglycemia - Weight gain - GI distress
Biguanides (Metformin) *first-line drug	decrease hepatic production of glucose and reduce insulin resistance	- GI effects: nausea, vomiting, abdominal discomfort, metallic taste, diarrhea - Headache, dizziness, anorexia, fatigue
Alpha-Glucosidase Inhibitors	Block enzymes that break down complex carbohydrates	- Abdominal cramping - Diarrhea - Flatulence

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INTERVENTIONS- ANTIDIABETIC DRUGS

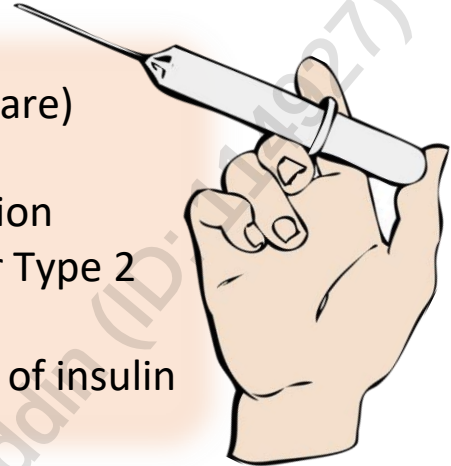
DRUG NAME	MECHANISM	SIDE EFFECTS
Meglitinides	stimulate insulin release from pancreatic islet cells	- Hypoglycemia - Weight gain - Flu-like symptoms
Thiazolidinediones	improve insulin sensitivity, transport, and utilization	- Cardiovascular events - Weight gain - Edema
Incretin enhancers	Incretins signal the pancreas to increase insulin and the liver to stop glucagon production	- Flulike symptoms - URTI - Nausea, fatigue, headache, diarrhea

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MANAGEMENT OF CARE

- Assess skin integrity (foot and wound care)
- Monitor blood sugar
- Implement measures to prevent infection
- Client teaching on diet and exercise for Type 2
- Medication adherence
- Health teaching for self-administration of insulin



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CLIENT TEACHING – DURING ILLNESS

CLIENT EDUCATION DURING ILLNESS

Take medications as prescribed

Monitor blood sugar levels and test for urine ketones q 3 hours

If regular diet cannot be tolerated, have soft foods 6-8 times per day

Should vomiting or diarrhea occur, take liquids q 30-60 minutes to prevent dehydration



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CLIENT TEACHING – DURING ILLNESS



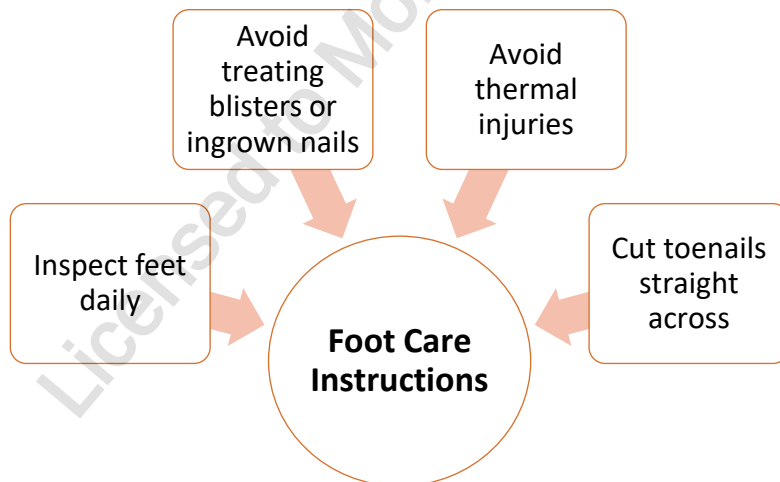
Notify HCP if:

- ✓ Vomiting, diarrhea, or fever persist
- ✓ Blood glucose increases to above 250-300 mg/dL
- ✓ Ketonuria is present for over 24 hours
- ✓ Unable to take in food or fluids for 4 hours
- ✓ Illness lasts longer than 2 days

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CLIENT TEACHING – FOOT CARE



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SUMMARY

- Type 1 Diabetes involves lack of insulin while Type 2 Diabetes involves resistance to insulin or defective insulin receptors
- Alpha and Beta islet cells of pancreas are responsible for production of insulin
- DKA often occurs in Type 1 Diabetics while HHNS often occurs in Type 2 Diabetics
- Insulin therapy is indicated for type 1 diabetics and type 2 diabetics that do not respond to other treatments
- Several classes of oral antidiabetic agents are indicated for use in type 2 diabetes
- Chronic complications of diabetes include retinopathy, neuropathy and nephropathy
- Client teaching must include foot care as well as guidelines during illness

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